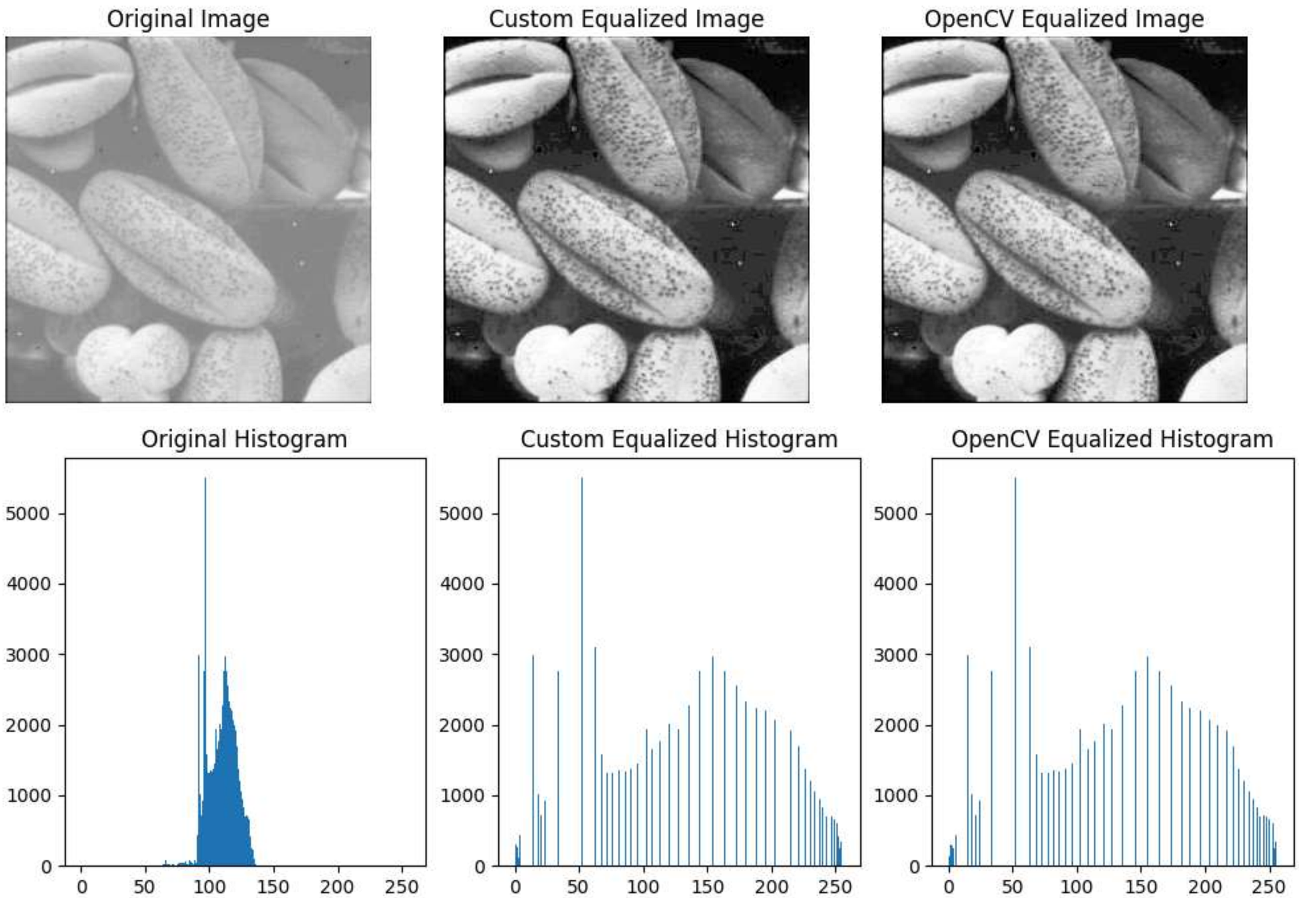
```python
import numpy as np
import cv2
import matplotlib.pyplot as plt

# ----------------------------------------------------------
# Custom Histogram Equalization Function
# ----------------------------------------------------------
def equalize_hist(image, L=256):
    intensity_levels, counts = np.unique(image, return_counts=True)
    total_pixels = image.size
    pdf = counts / total_pixels
    cdf = np.cumsum(pdf)
    transform = np.floor((L - 1) * cdf).astype(np.uint8)
    mapping = dict(zip(intensity_levels, transform))
    equalized_image = np.vectorize(mapping.get)(image)
    return equalized_image.astype(np.uint8)

# ----------------------------------------------------------
# Read Uploaded Image
# ----------------------------------------------------------
image = cv2.imread("example.jpg", cv2.IMREAD_GRAYSCALE)

if image is None:
    print("Error: Image not found. Make sure filename matches upload.")
else:
    # Apply histogram equalization
    custom_image = equalize_hist(image)
    opencv_image = cv2.equalizeHist(image)

    # ------------------------------------------------------
    # Display Images
    # ------------------------------------------------------
    plt.figure(figsize=(12, 4))

    plt.subplot(1, 3, 1)
    plt.imshow(image, cmap='gray')
    plt.title("Original Image")
    plt.axis("off")

    plt.subplot(1, 3, 2)
    plt.imshow(custom_image, cmap='gray')
    plt.title("Custom Equalized Image")
    plt.axis("off")

    plt.subplot(1, 3, 3)
    plt.imshow(opencv_image, cmap='gray')
    plt.title("OpenCV Equalized Image")
    plt.axis("off")

    plt.show()

    # ------------------------------------------------------
    # Display Histograms
    # ------------------------------------------------------
    plt.figure(figsize=(12, 4))

    plt.subplot(1, 3, 1)
    plt.hist(image.ravel(), bins=256, range=[0, 256])
    plt.title("Original Histogram")

    plt.subplot(1, 3, 2)
    plt.hist(custom_image.ravel(), bins=256, range=[0, 256])
    plt.title("Custom Equalized Histogram")

    plt.subplot(1, 3, 3)
    plt.hist(opencv_image.ravel(), bins=256, range=[0, 256])
    plt.title("OpenCV Equalized Histogram")

    plt.show()
```



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